CONSUMER LOAN PORTFOLIO ANALYSIS

Composition, Pricing & Loan Profile

Python-Based Data Understanding, Cleaning, EDA and Visualization

PROJECT DOCUMENTATION

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Project questions: How is the loan portfolio composed by status, purpose, grade and homeownership? How are loans priced across credit grades? How do loan amounts and debt-to-income ratios vary across purposes and statuses?

1. Executive Summary

This project examines a consumer-loan dataset to understand portfolio composition, loan characteristics, credit grades, pricing, and loan status. The workflow covers data understanding, targeted cleaning, exploratory data analysis (EDA), visualizations, and a consolidated Python script. The analysis is descriptive: it profiles the portfolio and compares segments, but it does not assess default risk, establish causation, or predict future outcomes.

The original dataset contains 10,000 rows and 56 columns. After removing one unnecessary index-like column, the cleaned dataset contains 10,000 rows and 55 columns. Missing values in joint-application fields are retained where they may indicate that those fields are not applicable to individual applications.

2. Project Overview

Item	Description
Project	Consumer Loan Portfolio Analysis: Composition, Pricing & Loan Profile
Domain	Consumer lending / BFSI
Tools	Python, pandas, NumPy, Matplotlib, Jupyter Notebook, VS Code
Scope	Data understanding, cleaning, EDA, visualization, reusable scripting
Out of scope	Credit-risk modelling, predictive default analysis, machine learning, and Power BI in the current scope

3. Project Questions and Objectives

The project is organized around three descriptive questions:

1. How is the loan portfolio composed by loan status, purpose, grade, and homeownership?
2. How are loans priced across credit grades?
3. How do loan amounts and debt-to-income (DTI) ratios vary across purposes and loan statuses?

Objectives:

- Understand the structure and quality of the dataset.
- Clean selected fields while preserving meaningful missing values.
- Explore loan status, purpose, loan amount, interest rate, DTI ratio, grade, and homeownership.
- Use visualizations to communicate selected portfolio patterns.
- Document evidence, implications, and limitations without overstating conclusions.

4. Dataset Description

Dataset stage	Rows	Columns	Notes
Original	10,000	56	Raw source file; preserved unchanged.
Cleaned	10,000	55	Unnecessary Unnamed: 0 column removed.

The dataset contains borrower attributes, credit-history fields, loan terms and amounts, grades, loan statuses, and repayment-related measures. Exact field meanings should be verified against the source data dictionary, if available.

5. Tools and Technologies

- **Python**: core programming language for analysis.
- **pandas**: data loading, cleaning, grouping, and summary statistics.
- **NumPy**: numerical computing support.
- **Matplotlib**: static charts.
- **Jupyter Notebook**: stepwise exploration and documentation of analytical decisions.
- **VS Code**: consolidated, reusable Python script.

6. Methodology and Workflow

1. **Data understanding:** inspect columns, dimensions, data types, missing values, duplicates, categorical counts, and descriptive statistics.
2. **Data cleaning:** remove the unnecessary index-like column, apply targeted missing-value handling, standardize selected text, and convert issue_month to datetime.
3. **EDA:** compare loan status, purpose, amounts, interest rates, DTI ratios, grades, and homeownership.
4. **Visualization:** create four Matplotlib charts to communicate selected distributions and comparisons.
5. **Scripting:** consolidate the workflow into one Python script for repeatable execution.

7. Data Understanding

Initial inspection covered column names, dataset shape, data types, first rows, missing-value counts, duplicate rows, categorical distributions, descriptive statistics, and loan-status percentages. This established a baseline before transformations.

8. Data Cleaning Decisions and Justifications

Field / operation	Action	Reason / caution
Unnamed: 0	Removed	Unnecessary index-like column.
emp_title	Fill missing with Unknown; trim whitespace	Makes missing titles explicit and standardizes text.
emp_length	Fill missing values with median	Targeted numeric imputation; introduces estimated values.
num_accounts_120d_past_due	Fill missing with 0	Assumes missing means zero; verify source semantics.
debt_to_income	Fill missing values with median	Targeted imputation; may influence group averages.
Delinquency / inquiry recency fields	Fill missing values with 0	Only valid if source definitions support interpreting missing as none.
issue_month	Convert to datetime using %b-%y	Supports date-aware handling.
Joint-application fields	Retain missing values	May be not applicable to individual applications; avoids inventing values.
Duplicates	Checked after cleaning; 0 found	No exact duplicate rows detected.

Cleaning caution: imputing a missing recency or delinquency value with zero is semantically appropriate only if the dataset documentation supports that interpretation. These choices are described transparently.

9. Exploratory Data Analysis

The EDA examined:

- Loan status and loan purpose frequency distributions.
- Loan amount summary statistics and average amounts by purpose and status.
- Interest-rate summary statistics and average interest rate by grade.
- Average debt-to-income ratio by loan status.
- Loan status distributions across grades and homeownership categories.

10. Key Findings and Supporting Evidence

Finding	Evidence	Business implication	Limitation
Portfolio is dominated by Current loans.	9,375 of 10,000 loans are Current (93.75%).	Most records are in Current status at the observed snapshot.	Reflects status at a point in time; not a default or performance rate.
Debt consolidation is the most common purpose.	5,144 loans.	A major concentration in the observed portfolio.	Frequency alone does not indicate profitability or risk.
Interest rates generally rise from Grade A toward Grade G.	Grouped mean interest rates show an upward pattern.	Consistent with grade-level pricing differentiation.	Descriptive pattern; does not establish cause.
Average loan amount varies by purpose and status.	Group means calculated in EDA.	Shows where lending exposure is concentrated across segments.	Descriptive comparison only.
Average DTI differs by loan status.	Group means calculated in EDA.	A starting point for further investigation.	Averages alone do not establish predictive power or causation.

Rare-status note: only 7 loans are classified as Charged Off in the supplied counts. This sample is too small for meaningful analysis of that outcome, which is why this project profiles the portfolio rather than assessing credit risk. Sample sizes should be shown when discussing rare statuses.

11. Visualizations

Chart	Purpose
Loan Status Distribution	Shows the number of loans in each status category.
Loan Purpose Distribution	Shows which purposes occur most frequently.
Average Interest Rate by Loan Grade	Compares average interest rates across grades.
Average Loan Amount by Purpose	Compares average loan amounts across purposes.

Charts should be interpreted alongside category counts and data limitations, especially where sample sizes are small.

12. Business Implications

- Loan-status and loan-purpose distributions give a clear view of portfolio composition and concentration, for example in debt consolidation.
- Grade-level interest-rate differences are consistent with pricing differentiation across credit grades.
- Differences in loan amounts across purposes indicate where lending exposure is concentrated.
- Differences in DTI across statuses are a starting point for further investigation, not evidence of default risk.

13. Limitations and Analytical Cautions

- This is descriptive analysis; it does not assess credit risk, predict defaults, or establish causal relationships.
- The dataset contains very few non-Current outcomes (only 7 Charged Off records), which does not support a credit-risk analysis.
- Imputation decisions can influence statistics and should be documented.
- Joint-application fields retain missing values because they may not apply to individual applications.
- Dataset time coverage, sampling method, and field definitions should be verified before making broader claims.
- The Current-loan share must not be interpreted as a default rate or as proof of low risk.

14. Conclusion

The project establishes a reproducible descriptive workflow for profiling a consumer-loan portfolio. It summarizes the dominant loan status and purpose, examines differences in loan amounts and interest rates across segments, and compares debt-to-income ratios across statuses. The results describe the portfolio and provide a basis for further work. Because the dataset has very few non-Current outcomes, a credit-risk analysis would require a richer set of repayment outcomes.

15. Project Structure and Reproducibility

Repository structure:

```
consumer-loan-portfolio-analysis/
|-- Documentation/
|   `-- Project_documentation.pdf
|-- Scripts/
|   `-- loan_analysis.py          # consolidated analysis script
|-- data/
|   |-- Loans.csv                 # raw source; keep unchanged
|   `-- Loans_Cleaned.csv         # exported after cleaning
|-- notebooks/
|   |-- 01_Data-Understanding.ipynb
|   |-- 02_Data-Cleaning.ipynb
|   |-- 03_EDA.ipynb
|   `-- 04_Visualization.ipynb
`-- README.md
```